INCH-POUND

MIL-C-0040153K(GL)
7 January 1991
USED IN LIEU OF
MIL-C-40153J
12 March 1984

MILITARY SPECIFICATION

CLOTHING REPAIR SHOP, TRAILER MOUNTED

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-40153J. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers a self-contained clothing repair shop, designed to be transported by and removable from a standard Government furnished trailer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Documents Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3530

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SPECIFICATIONS

FEDERAL

A-A-632	-	Flatirons, Electric
A-A-2344	-	Wrenches, Adjustable
A-A-50470	-	Oilers, Hand, Filler
P-D-680	-	Dry Cleaning and Degreasing Solvent
OO-S-256/GEN	-	Sewing Machine, Industrial; General Specification For
OO-S-256/1	-	Sewing Machines, Industrial; Flatbed, Single-Needle, Machines
OO-S-256/3	-	Sewing Machines, Industrial; Cylinder Bed, Single-Needle, Machines
OO-S-256/8	-	Sewing Machines, Industrial; Button-Sewing Machine
RR-I-660	-	Ironing Board, Pads and Covers; (Household)
TT-P-664	-	Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, VOC-Compliant
GGG-P-471	-	Pliers; Pliers, Slip Joint
GGG-S-121	-	Screwdriver and Screw Starter, Hand
GGG-S-278	-	Shears and Scissors
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-P-40	-	Packaging and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

MILITARY

MIL-P-116	-	Preservation, Methods Of
MIL-B-117	-	Bags, Sleeves and Tubing
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank
MIL-P-2001	-	Punch, Cutting, Revolving Head
MIL-M-7298	-	Manuals, Commercial Off-the-Shelf
MIL-T-10579	-	Trailers, Cargo: 2 Wheel, 1/4 Ton to 1-1/2 Ton
MIL-S-85634	-	Sealer, Heat, Seam Tape and Patch, Anti-Exposure Coverall

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-642	-	Identification Marking of Combat and Tactical Transport Vehicles
MS 35914	-	Insert, Screw Thread - Thread Cutting

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FEDERAL

FED-STD-595 - Colors Used in Government Procurement

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

6-1-9557 - Clothing Repair Shop, Trailer Assembly and List of Drawings
6-1-9558 - Clothing Repair Shop Working Configuration
6-1-6384 - Clothing Sewing Machine Assembly
6-1-6385 - Darning Sewing Machine Assembly
6-1-6388 - Button Sewing Machine Assembly
6-1-9783 - Stand Assembly
6-1-1117 - Table Folding

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the issue of the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN WELDING SOCIETY (AWS)

Standard Qualification Procedure of the American Welding Society
A5.10 - Aluminum and Aluminum Alloy Bare Welding Rods and Electrodes

(Application for copies should be addressed to the American Welding Society, 550 N.W. Le Jeune Road, P.O. Box 33126, Miami, FL 33135.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electric Code

(Application for copies should be addressed to National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269.)

NATIONAL ELECTRIC MANUFACTURERS ASSOCIATION (NEMA)

MG 1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

ASSOCATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open-Top Cars

(Application for copies should be addressed to the Assocation of American Railroads, 50 F Street NW, 5th Floor, Washington, DC 20001.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Alternate components. Components offered as equivalent to components specified herein and on the reference drawings as a specific manufacturer's part number shall be functionally equal with the manufacturer's part number identified. The incorporation of such a component in the design of the specified end product shall not require modification to any other component, unless such modification is specifically approved by the contracting officer. Prior to manufacture of the first article sample, if none is required, prior to commencing production, the contractor shall submit for the contracting officers' approval, a list identifying each alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed

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substitution. Approval of the submitted listing, together with necessary supporting data does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials and components. The materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document.

3.3.1 Lumber. Lumber used in the fabrication of the sewing machine trays, table tops, and wood bases for folding stands and tables, shall be uniformly kiln dried without brashness, case-hardening, or honeycombing, in such a manner as to "firmly set" natural pitch and resin. Moisture content of the lumber at time of assembly shall be 8 to 12 percent. Lumber shall be further selected to eliminate such defects as knots exceeding 1/4-inch average diameter, decay, mineral streaks, wane, shake, and splits. Any one of the following species of wood shall be used:

Ash	Maple, hard or soft
Beech	Hickory
Birch, yellow or sweet	Oak, red or white

3.4 Design and construction. Design and construction of the clothing repair shop shall be as specified herein and in accordance with Drawing 6-1-9557 and all drawings listed thereon.

3.4.1 Electrical. All wiring shall be done in accordance with the provisions of NFPA No. 70. All wiring devices shall be UL listed. All motors shall be constructed, classified and rated in accordance with NEMA MG 1.

3.4.2 Fabricating general. The cabinet frame and racks are welded construction. The sheathing, towing skids, hardware, and lifting eye assemblies are noted on the outermost sections only. The outermost pieces may be set in place and used as templates to locate the underlying holes. Contractors who prelocate all holes are responsible for their correct alignment.

3.4.3 Sewing machine assemblies. The sewing machine assemblies shall conform to OO-S-256 and the applicable slash sheets, and the requirements of detailed paragraphs 3.4.3.1 thru 3.4.3.3 of this specification. When in conflict with OO-S-256, this specification shall take precedence. The requirements for the stands and table tops shall be as noted in paragraphs 3.4.3.4 and 3.4.3.5 and as specified on the drawings. The overall dimensions of the sewing machines and related components shall be such as to fit within the spaces provided in the

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cabinet as shown on the drawings. In addition to the supplies called for in the detailed paragraphs, each sewing machine shall be supplied with one each of the following conforming to OO-S-256:

Bobbin winder	Operators manual
2 spool thread unwinders	Maintenance manual
Oil drip pan	Parts manual
Size 3L belts (length determined by manufacturers mounting)	Lubrication chart

The manuals shall be prepared in accordance with MIL-M-7298. The bobbin winder and thread unwinders stand shall be mounted to the table top with corrosion resistant steel (CRES) screws and CRES inserts (inserts per MS 35914). The screw and insert sizes shall be compatible with each other and with the hole sizes of the thread stand and bobbin winder. The thread unwinder base and bobbin winder mounting hole patterns shall be used as a pattern for mounting the inserts into the table top. The machine motors and lamps shall be as specified on the applicable drawings.

3.4.3.1 Machine, clothing. The clothing machine shall be a light-medium duty machine conforming to OO-S-256/1, class 2, stitch type 301. The bed size shall be 18-3/4 by 7 inches. Each machine shall be provided with two hinge plates appropriate for mounting the machine head to the table top. Each machine shall be furnished with a round bar treadle, treadle retainer clips, treadle extension and ball joint linkage rod. All machine heads, stands, table tops, motors and accessories shall be assembled per Drawing 6-1-6384. The hinge plates shall be mounted per the machine head manufacturer's instructions. Each machine shall be supplied with the following spare parts:

Item identification	Quantity
Presser bar	1
Needle bar thread guide	1
Needle bar thread guide screw	1
Needle clamp, complete	2
Throat plate	1
Throat plate screw	2
Feed dog screw	2
Tension assembly, complete	1
Face plate screw	1
Face plate thumb screw	1
Presser bar lifting lever	1
Bed slide plate spring screw	1
Thread guide set (4 pc)	1
Hinged presser foot	1
Bobbin case	1
Bobbin, metal	10
Shuttle and race assembly	1
Thread take-up spring	1
Shuttle race screw	2

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Needle, round point, chrome finish:

Metric sizes	
110	100
120	100
130	100
140	100

3.4.3.2 Machine, darning. The darning machine shall conform to OO-S-256/3, class 3. Each machine shall be supplied with 100 round point chrome finish needles, sizes 16 and 18, 15 metal bobbins, 5 bobbin cases and a vertical access rotary hook. Each machine shall be furnished with two round bar foot pedals with retainer slips and springs to normally return to the up position, and power and foot lifter chains. All machine heads, stands, table tops, motors and accessories shall be mounted per Drawing 6-1-6385.

3.4.3.3 Machine, button. The button machine shall conform to OO-S-256/8, class 1. It shall have a high impact plastic eye guard. The machine shall be provided with power and foot lifter chains. Each machine shall be furnished with two round bar foot pedals with retainer clips and springs to normally return to the up position. All machine heads, stands, table tops, motors and accessories shall be mounted per Drawing 6-1-6388. The button machine shall be suitable for sewing two and four hole flat buttons from 20 to 45 lines close to or away from light to heavyweight materials. It shall be a single needle, 101 stitch machine with a vibrating needle bar and a cylinder bed. It shall be designed for sewing on buttons with 16 stitches including cross over and knotting stitches. The machine shall operate at 1000 stitches per minute. The machine shall be furnished with an attachment for stay buttons, and accessories for shank buttons. One hundred each stub point chrome finish spare needles, sizes 16 and 18, a set of loopers, and 6 needle threaders shall be furnished.

3.4.3.4 Sewing machine stands. The sewing machine stands shall be designed in accordance with Drawing 6-1-9783 and all associated drawings.

3.4.3.5 Sewing machine table tops. The sewing machine table tops shall be constructed out of solid wood sections laminated together. Wood sections shall conform to 3.3.1. The outside dimensions of the table tops are noted on the drawings. The configuration for the openings shall conform to the machine head manufacturer's requirements. In addition, they shall be appropriate for their intended use. The upper and lower outer perimeter edges shall have a quarter round section removed. The quarter round section shall have a minimum radius of 0.5 inches. The tops shall be joined, planed and sanded to a flat smooth surface. All surfaces shall be finished in accordance with the applicable drawings. All hardware and accessories required to mount the components shall be provided. The stands shall be connected to the tables using 0.32 by 2.50 inch CRES dome head carriage bolts. Bolts shall be fastened through the stand and table top using CRES washers and nuts. The machine motors shall be mounted using CRES washers, nuts and dome head carriage bolts.

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3.4.3.6 Sewing machine tray assemblies. The sewing machine tray assemblies shall supply, support and provide anchorage for the sewing machine heads during transport. They shall also be suitable for safe personnel handling during machine head movement from the cabinet to the work place. The broad characteristics for the three types of trays required shall be as specified herein and as shown on the drawings. The actual details for the tray assemblies will be dependent on the machine heads. The tray assemblies shall be finished in accordance with the applicable drawings. The sewing machine tray assemblies shall include but not be limited to the following items:

Base plate
Machine base with bed frame
Blocking
Brass wood screws
Handles
Tie down strap assembly
Felt cushions
Identification plate
Hardware

3.4.4 Hand tack, button attaching machine. The button attaching machine shall be furnished with dies as shown in table I. Commercial plated standard nuts, bolts, and washers shall be furnished in the base of the button attaching machine for mounting the machine on the folding leg table.

TABLE I. Button attaching machine dies

Machine	Item identification	Quantity
Button attaching	Upper closed	
	Die, Sure Snap, part no. 3032	1
	Upper open	
	Die, Sure Snap, part no. 2727	1
	Lower Die, Sure Snap, part no. 312	1
	Lower Die, Sure Snap, part no. 351	1

3.4.5 Iron, hand electric. The electric hand iron shall conform to type II of A-A-632.

3.4.6 Ironing board, pad and cover. The ironing board shall conform to type I of RR-I-660, except that the top dimensions shall be 27 by 7 inches overall, and the stand shall not be adjustable and shall be suitable for mounting on the folding table, per Drawing 6-1-1117. The pad shall be type III, class A, and the cover shall be type IV, class C conforming to RR-I-660 and shall fit the ironing board top.

3.4.7 Press, chucks and dies. The chucks and dies for the grommet press shall be furnished as shown in table II.

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TABLE II. Grommet press chucks and dies

Machine	Item identification	Quantity
Grommet press	Chuck, United Carr, part no. 1486	1
	Chuck, United Carr, part no. 1580	1
	Chuck, United Carr, part no. 9219R	1
	Chuck, United Carr, part no. 9323	1
	Chuck, United Carr, part no. 9447	1
	Chuck, United Carr, part no. 9470	1
	Die, United Carr, part no. 1483	1
	Die, United Carr, part no. 1488	1
	Die, United Carr, part no. 1587	1
	Die, United Carr, part no. 9182	1
	Die, United Carr, part no. 9454	1
	Die, United Carr, part no. 9471	1

3.4.8 Machine, heat sealing. The heat sealing machine shall conform to MIL-S-85634 (see 6.6).

3.4.9 Components and accessories. Each clothing repair shop shall be furnished as specified in table III. Belting and hooks (items 1 and 2) shall be compatible with the machines furnished.

TABLE III. Clothing repair shop hand tools and accessories

Item	Description	Quantity
1	Belting, leather, round, 1/4-inch	50 feet
2	Hooks, belt	25
3	Oiler, hand, push-bottom, steel, 4-inch rigid spout, 1/2 pint, type I, class A or B of A-A-50470	4
4	Pliers, slip joint, straight nose, w/cutter, 8-inch length, type II, class 2, style A of GGG-P-471	1
5	Punch, cutting, revolving head, 6 tubes of MIL-P-2001	1
6	Screwdriver, flat tip, straight side, plastic handle, 9/64 by 3-inch blade length, type I, class 1, design A of GGG-S-121	1
7	Screwdriver, flat tip, straight side, plastic handle, 3/16 by 3-inch blade length, type I, class 1, design A of GGG-S-121	1
8	Screwdriver, flat tip, plastic handle, 9/64 by 1-1/2 inch blade, type I, class 1, design A of GGG-S-121	1

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TABLE III. Clothing repair shop hand tools and accessories (cont'd)

Item	Description	Quantity
9	Screwdriver, flat tip, plastic handle, 1/4 by 4-inch blade, type I, class 5, design A of GGG-S-121	1
10	Shears, 10 inch overall length, type I, class 2 of GGG-S-278	4
11	Thimble, sewing, brass, chrome plated, closed, large size, manufacturer's commercial standard	4
12	Thimble, sewing, brass, chrome plated, closed, medium size, manufacturer's commercial standard	4
13	Wrench, adjustable, crescent-type, 10-inch length, type I, class 1 of A-A-2344	1

3.5 Performance.

3.5.1 Over road and terrain. There shall be no evidence of damage to the cabinet or the components in the cabinet when subjected to the road and terrain test as specified in 4.5.1.

3.5.2 Rainproof. Each cabinet shall be rainproof, when tested as specified in 4.5.2.

3.5.3 Operational. When operationally tested as specified in 4.5.3, each coil of the generator set shall not exceed a current of 15 amps.

3.6 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.2.1).

3.7 Data name plates and marking. All data name plates shall be aluminum or brass. For Marine Corps use (see 6.2), a data name plate will be furnished by the Commandant of the Marine Corps (CSG), Headquarters US Marine Corps, Washington, DC 20380.

3.7.1 Marking for identification of the end item. In addition to other required marking, each cabinet for Army use (see 6.2) shall have stenciled or painted in black Gothic style lettering "US ARMY" 2-inches high, and directly under name "CLOTHING REPAIR SHOP" 1-inch high, centered on outside of the two sides and the rear wall. For Marine Corps use (see 6.2), the characters "USMC" shall be substituted for "US ARMY" and the color shall be semi-gloss yellow 23538 of FED-STD-595.

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3.7.2 Marking for identification of components and their location. Each sewing machine tray shall have a metal plate indicating the manufacturer's sewing machine head number. A metal plate, corresponding to the plate on each tray, shall be fastened to one end of the tabletop to which the machine is to be mounted, and another plate to the inside of the cabinet, to indicate the storage location of the tabletop. Metal plates shall be attached with corrosion-resisting metal screws or rivets. The plates shall be located so they are readily visible from the outside of the cabinet when the doors are open. Plates shall have markings in accordance with type III of MIL-P-514.

3.7.3 Registration number. The location of the lettering and numerals on the cabinet assembly, shall be in accordance with MIL-STD-642.

3.8 Government furnished and loaned property.

3.8.1 Trailer, cargo, 1-1/2 ton 2W M105A2. The trailer will be a 1-1/2 ton 2W M105A2 cargo trailer conforming to MIL-T-10579 and rear support leg conforming to Ordnance Corps Drawing No. 8681929 for Kit Support Leg and will be Government furnished property (see 6.2 and 6.4).

3.8.2 Generator, set(s), 3 kilowatt. The Government will loan a generator set(s) (3 kilowatt), for first article testing, when required (see 6.2 and 6.5), and for end item testing.

3.9 Workmanship. The end item shall conform to the quality of product established by this specification. The units shall be free from fractures, splits, punctures, dents, tears, creases, bows, miscasts, deteriorations or malfunctions. There shall be no sharp edges, burrs or slivers. The finish shall be free of blisters, peel, chipped areas or areas of no film. The finish shall be free from smudges, poor touch up, scale, scratches, stain, overspray sag, imprint or embedded foreign matter. Threaded fasteners shall not be broken, stripped or loose.

3.9.1 Welders. Before assigning any welder to work, the contractor shall provide the contracting officer with names of the welders to be employed in the work, together with certification that each of these welders has passed a qualification test as prescribed by either of the following listed codes for the type welding operation to be performed, and that such qualification is effective as defined by that particular code as follows:

a. Standard Qualification Procedure of the American Welding Society
b. Welding Qualification of the American Society of Mechanical Engineers

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.2 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.5. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component testing. The electrical and mechanical components for each production end item shall be tested in accordance with 4.5.3, except as noted. The operational test is a test of the equipment and electrical distribution system. It may be done independently of the cabinet fabrication. Commercial three phase, 208 V, 60 cycle, power may be substituted for the Government loaned

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generator. Upon successful compliance with 3.5.3 each equipment group and electrical distribution system shall be tagged inspected, and may be preservation packaged. Failure to pass three tests shall be cause for rejection of the equipment and distribution system.

4.4.1.2 Cabinet assembly test. Each cabinet assembly shall be tested in accordance with 4.5.2. Failure to meet the requirement in 3.5.2 shall be cause for rejection of the cabinet assembly.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.3 End item visual examination. Each fully assembled end item shall be examined for completeness of assembly, presence and adequacy of components, specified types of components, workmanship and markings. The lot size shall be expressed in units of completely assembled clothing repair shops. The sample unit shall be one completely assembled clothing repair shop. Any deviation from specified requirements shall be cause for rejection of the end item.

4.4.4 End item dimensional examination. Each fully assembled end item shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. The lot size shall be expressed in units of completely assembled clothing repair shops. The sample unit shall be one completely assembled clothing repair shop. Any dimension not within the specified tolerances shall be cause for rejection of the end item.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

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4.5 Methods of inspection.

4.5.1 Road and terrain tests. The end item shall be towed fully loaded over not less than 5 miles of paved roads at speeds up to fifty (50) miles per hour (mph), as road and weather conditions permit; over not less than 5 miles of graveled roads at speeds averaging thirty-five (35) mph and over not less than five (5) miles of open terrain and cross country at speeds averaging 5 mph to determine compliance with 3.5.1.

4.5.1.1 Paved and graveled roads. The paved and graveled road tests of 4.5.1 shall each include not less than 20 quick starts and stops on grades of not less than 5 percent or greater than 10 percent. Fifty percent of the starts and stops shall be made on ascending grades and fifty percent on descending grades.

4.5.1.2 Open terrain and cross country. The open terrain and cross country course of 4.5.1 shall have surface irregularities such as pot holes (averaging not less than 4 inches in depth), mounts, rocks and brush, randomly spaced more than 10 feet apart. The test drive shall include ten abrupt starts, ten sharp left hand turns, and ten sharp right hand turns.

4.5.2 Rain test. Upon completion of the road and terrain test, the end item shall be exposed to simulated rainfall at a rate of 3 inches, plus or minus 1 inch, per hour as measured at the surface of the equipment by a US Weather Bureau type gauge. The direction of rainfall shall include all angles from vertical to 45 degrees from vertical. The rainfall shall be dispersed uniformly over the test area within the rainfall limits specified above. Each of the four sides, individually or simultaneously, shall be exposed to the rainfall for not less than 15 minutes. Each side exposure shall include exposure of the top. Examination shall be made to determine compliance with the requirement in 3.5.2.

4.5.3 Operational test. The electrical and mechanical components shall be set up for operation as noted below. For the first article test, the operational test shall be done upon completion of the rain test (see 4.5.2). The ambient temperature shall not be less than 40°F.

a. All equipment shall be set up for operation in an arrangement as indicated on Drawing 6-1-9558 with an industrial analyzer properly installed in the cable leaving the Government loaned generator.

b. All sewing machines shall be completely assembled and properly lubricated in accordance with the manufacturer's instructions.

c. All starting switches shall be in the "OFF" position.

d. Sewing machine needles and bobbins shall be removed from the darner, button, and clothing machines.

e. The generator shall be started and adjusted for 208 volts, 60 cycles three phase electrical output and 110 volts, 60 cycles, single phase electrical output.

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f. After the generator has warmed up, the generator breaker shall be placed in the "ON" position.

g. Check to make sure all extensions and work lights are connected.

h. Start all sewing machines and turn on all machine and work lights.

i. Apply load to all sewing machines by depressing the operating treadle. Load shall be applied for a short period only in order to obtain electrical readings on the industrial analyzer. This period shall not exceed more than 4 minutes for any single run.

j. Record the voltage and current for each power line.

k. Repeat h. and i. two more times and record the average reading for all three runs.

l. Using the data recorded above, the electrical load on each generator coil shall be calculated using the following formula to determine compliance with the requirement in 3.5.3:

$$\text{Approximate load on each coil} = \frac{I}{\sqrt{3}}$$

Where: I = Current (amperes) in power line A, B or C

4.5.4 Sewing machine test. Upon completion of the operational test (see 4.5.3), the first article sewing machines shall be tested as sepcified in OO-S-256 using the button, darning, and one of the clothing machines. Failure to meet the requirements specified in 3.4.3 shall result in rejection of the first article.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Trailer chassis. The trailer chassis shall be preserved in accordance with MIL-T-10579.

5.1.1.2 Sewing machine heads. The head shall be removed from each sewing machine and the exterior, uncoated metal surfaces cleaned with solvent conforming to P-D-680. Exterior surfaces, cleaned as specified, and all interior surfaces of each machine head, except sealed oil reservoirs, shall be thoroughly coated with type P-7 preservative of MIL-P-116. Special care shall be exercised to insure that all interior surfaces specified are coated with preservative. All sharp projecting edges on the machine shall be covered with

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minimum 1/4-inch thick cushioning material conforming to type II, class B of PPP-C-843, and the cushioning material secured in place with cord or twine. Cushioning material placed against preserved surfaces shall be prevented from contacting the preservative by the use of barrier material conforming to type II, grade A of MIL-B-121. Each machine head shall then be placed in a bag conforming to type I, class E of MIL-B-117 and the bag closed by heat sealing. The included air volume of the bag shall be kept to a practicable minimum. Machine heads shall be placed in the appropriate tray, secured with straps provided, and the tray secured in the trailer.

5.1.1.3 Hand tack button attaching machine and hand grommet press. All uncoated ferrous metal surfaces of each hand tack button attaching machine and hand grommet press shall be cleaned by process C-1 of MIL-P-116 and thoroughly dried. Surfaces cleaned as specified shall be thoroughly coated with type P-7 preservative of MIL-P-116. The hand tack button attaching machine and the hand grommet press shall then be individually prepared by method IA-14 of MIL-P-116. The preserved machine and press shall be placed in the stowage box on the trailer and secured in such a manner as to prevent movement while in transit.

5.1.1.4 Remaining components.

5.1.1.4.1 Cleaning and drying. all exposed, uncoated, ferrous metal surfaces of the remaining components of each clothing repair shop shall be cleaned by process C-1 of MIL-P-116 and thoroughly dried.

5.1.1.4.2 Application of preservation. Surfaces, cleaned as specified shall be coated with preservative. Preservative shall be type P-2 for noncontacting surfaces (surfaces that do not contact other surfaces in operation) and type P-7 of MIL-P-116 for contacting surfaces (surfaces that contact other surfaces in operation).

5.1.1.4.3 Pulleys. Belts shall be removed from ferrous metal pulleys and the pulleys coated with a primer conforming to TT-P-664.

5.1.1.4.4 Securing. Components of each clothing repair shop, such as the stands, tables, chairs, etc., shall be secured with means provided.

5.1.1.4.5 Trailer body. The doors of the trailer body shall be secured in the closed position and padlocks locked in place. Padlock keys shall be securely wire-tied to the lock or lock staple with a minimum 0.057-inch diameter annealed wire.

5.1.1.4.6 Hand tools and accessories. Unless otherwise specified in the applicable item specification, the hand tools and accessories shall be preserved in accordance with PPP-P-40 except that items not covered therein shall be individually preserved in sealed clear polyethylene film bags and shall be protected in accordance with method III of MIL-P-116. The quantities per unit pack shall be limited to those as listed in 3.4.3.1, 3.4.3.2, 3.4.3.3 and 3.4.9. The unit packed tools and accessories units shall be placed in the appropriate drawer or location of the shop.

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5.1.2 Commercial preservation.

5.1.2.1 Trailer chassis. The trailer chassis shall be preserved in accordance with MIL-T-10579.

5.1.2.2 Sewing machine heads. The head shall be removed from each sewing machine and preserved in accordance with ASTM D 3951. Machine heads shall be placed in the appropriate tray, secured with the straps provided, and the tray secured in the trailer.

5.1.2.3 Hand tack button attaching machine and hand grommet press. Hand tack button attaching machine and hand grommet press shall be preserved in accordance with the industry's practice and individually unit packed in a snug-fitting fiberboard box conforming to any style, type CF or SF, class domestic of PPP-B-636. The unit packed machine and press shall be placed in the stowage box on the trailer and secured in such a manner as to prevent movement while in transit.

5.1.2.4 Remaining components.

5.1.2.4.1 Preservation. The remaining components of the repair shop shall be preserved in accordance with the industry's practice.

5.1.2.4.2 Securing. Components of each repair shop, such as the stands, tables, chairs, etc., shall be secured with means provided.

5.1.2.4.3 Trailer body. The doors of the trailer body shall be secured in the closed position and padlocks locked in place. Padlock keys shall be securely wire-tied to the lock or lock staple with a minimum 0.057 diameter annealed wire.

5.2 Packing. Packing shall be level A.

5.2.1 Level A packing.

5.2.1.1 Railway shipment. Loading of vehicles on open top cars shall be in accordance with the Association of American Railroads rules governing the loading of commodities on open top cars, section 6. Tire pressure shall be increased ten pounds over the required running pressure. The parking brake shall be set and locked and the levers wired or blocked in position.

5.2.1.2 Highway haulway shipment. The clothing repair shop shall be placed on bed or rails of haulway vehicles so that the load is evenly distributed and secured. Loading of units for haulway shall be in accordance with the Department of Transportation Motor Carrier Safety Regulations. Tire pressure shall be increased ten pounds over the required running pressure. The parking brake shall be set and locked and the levers wired or blocked in position.

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5.2.1.3 Highway towaway shipment. Towaway rules for shipment by driveaway shall be in accordance with the Department of Transportation Motor Carriers Safety Regulations. No processing of the chassis running gear shall be performed prior to highway shipment that would impair necessary operation or safety while in transit. Processing of the chassis running gear, not accomplished prior to highway shipment, shall be performed at intermediate destination when vehicles are to be placed in storage or shipped overseas. Tire pressure shall be the required running pressure.

5.3 Marking. In addition to any special marking required by the contract or purchase order, the clothing repair shop, components and accessories, unit and intermediate packs shall be marked in accordance with MIL-STD-129, or ASTM D 3951, as applicable. Special requirements applicable to unpacked vehicles shall apply.

5.3.1 Humping precautions. A large sign containing the words "DO NOT HUMP" shall be secured in a conspicuous place on each side of railroad cars used to ship the repair shops.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The trailer mounted clothing repair shop is intended for repairing the clothing of troops in the field.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Contract data requirements (see 3.6 and 6.2.1).
e. When data name plate is for Marine Corps (see 3.7).
f. Whether Army or Marine Corps marking is required (see 3.7.1).
g. Source and delivery of Government furnished property (see 3.8.1 and 6.4).
h. Source and delivery of Government loaned property (see 3.8.2 and 6.5).
i. Levels of preservation and packing (see 5.1 and 5.2).

6.2.1 Contract data requirements. Any data items to be delivered under any contract covered by this document shall be specifically called for in the contract in accordance with the applicable regulation of the contracting activity (see 3.6 and 6.2).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Government furnished property. The contracting officer should arrange to furnish the property specified in 3.8.1 and 6.2.

6.5 Government loaned property. The contracting officer should arrange to loan a generator set(s) (3 kilowatt - NSN 6115-00-017-8237), specified in 3.8.2 and 6.2.

6.6 Machine, heat sealing. The heat sealing machine is available from George Knight Co., 54 Lincoln Street, Brockton, MA 02403 or W.L. Gore, 3 Blue Ball Rd., P.O. Box 1130, Elkton, MD 21921.

6.7 Subject term (key word) listing.

Darning
Field
Sewing

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - MC

Review activity:

DLA - GS

Preparing activity:

Army - GL

(Project 3530-A088)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-0040153K(GL)	1991 January 7

3. DOCUMENT TITLE
CLOTHING REPAIR SHOP, TRAILER MOUNTED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4531	(2) AUTOVON 256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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